For a diatomic molecule (at high temperature), f = 7

For a triatomic molecule (linear type), f = 7

For a triatomic molecule (non-linear type), f = 6

- The average kinetic energy of sample of gas is equally distributed to all the degrees of freedom. It is called law of equipartition of energy. Each degree of freedom will get $\frac{1}{2}kT$ energy.
- The ratio of molar specific heat at constant pressure and constant volume of a gas $\left[\gamma = \frac{C_p}{C_v}\right]$

For

Monoatomic molecule: 1.67

Diatomic molecule (Normal temperature) : 1.40

Diatomic molecule (High temperature): 1.28

Triatomic molecule (Linear type): 1.28.

Triatomic molecule (Non-linear type): 1.33

- The mean free path $\lambda = \frac{kT}{\sqrt{2}\pi d^2 P}$. The mean free path is directly proportional to temperature and inversely proportional to size of the molecule and pressure of the molecule
- The Brownian motion explained by Albert Einstein is based on kinetic theory. It proves the reality of atoms and molecules.

UNIT-9(XI-Physics_Vol-2).indd 181 26-03-2022 17:47:46